

AS01 -- Corpus Construction and Characterization

AI-assisted immigration & refugee law case research | Nishanth Madthila | 95-820 Applications of NL(X) and LLM | Fall 2026

1. Overview

This memo reports on a 220-record corpus of Australian Federal Court opinions naming the Minister for Immigration as a party -- protection/refugee visa appeals, visa cancellations, and related migration matters decided 2006-2009. The corpus is my individual subtopic within our group's shared broad topic, legal case research. Source records come from the UCI Machine Learning Repository's "Legal Case Reports" dataset (Galgani, Compton & Hoffmann), itself drawn from AustLII, filtered to the immigration subtopic by a case-name match against "Minister for Immigration" (917 of 3,890 total cases matched; 220 were sampled at random, seed 42).

Three fields are extracted from each opinion, defined in **taxonomy.json**: **visa_category** (open label, no ground truth in the source data), **citation_treatment** (an 11-value enum copied from the source dataset's own citation-class annotation where available), and **cited_cases** (a list of case names/citations, copied exactly as written). The 25-record evaluation sample in **human_labels.jsonl** was hand-labelled by me directly from each opinion's text and, where present, the dataset's own citation table.

2. Human vs. Phi-4-mini-instruct (30%)

Phi-4-mini-instruct was run locally (Apple Silicon / mps, greedy decoding) on all 25 evaluation records using the baseline prompt (**v1_original**), which shows the model the JSON Schema built from taxonomy.json and asks it to "respond with a single JSON object matching this schema."

Metric	Value
Records evaluated	25
All-fields-correct rate	0.0%
Schema-violation rate	88% (22 of 25)
visa_category accuracy	4%
citation_treatment accuracy	0%
cited_cases accuracy (exact set match)	12%
cited_cases precision / recall	1.00 / 0.022

Root cause. Inspecting raw model output (not just the parsed/violation flags) showed the near-total failure was a format problem, not a content problem. In roughly 88% of records, Phi echoed the JSON Schema envelope it was shown -- a **{type, properties, required}** wrapper -- back as its answer, with the actual field values nested one level down inside **properties**, instead of returning a flat object at the top level. For example, on legal_08_1156 Phi returned:

```
{"type": "object", "properties": {"visa_category": "Skilled Independent Overseas Student (Class DD, subclass 880)", "citation_treatment": "considered", "cited_cases": [...] }, "required": [...] }
```

The content inside **properties** was frequently substantively reasonable (here, a correct read of the visa type) -- the model understood the case, it simply mirrored the shape of the instruction it was given rather than producing data in that shape. Because **validate_against_schema** checks for a top-level **visa_category** key, every one of these was flagged **\$.visa_category: missing required key**, which explains both the 88% schema-violation rate and the near-zero field accuracy in one mechanism.

3. Recovery Strategy (15%)

The starter code's default **v2_recovery** prompt only tightens enum-adherence wording ("MUST be exactly one of...") and asks the model not to explain itself -- it does not address the schema-echo failure identified above. Rerunning the 25 originally-failed records with that default v2 confirmed it does not help: the schema-violation rate stayed effectively at 100% and accuracy did not improve. This negative result confirmed the diagnosis needed to be more targeted than a generic "try harder" prompt.

I rewrote **v2_recovery** to name the actual failure directly: it tells the model explicitly that the JSON Schema shown is a description of the answer's shape, not a template to copy, shows a concrete example of a correctly-shaped flat response, and explicitly forbids the **type/properties/required** keys and markdown code fences. Rerunning the originally-failed records with this revised prompt:

Field	v1 (before)	v2 (after)	Δ
visa_category accuracy	4%	76%	+72 pp
citation_treatment accuracy	0%	48%	+48 pp
cited_cases accuracy (exact set)	12%	24%	+12 pp
cited_cases recall	0.022	0.162	7x
cited_cases precision	1.00	0.688	-0.31
Schema violations (of 25)	22	0	-22
Fully-correct records	0	2	+2

Fixing the output-shape bug recovered most of the signal: 2 records became fully correct end to end (**legal_06_78**, **legal_06_1495**), and every field's accuracy improved substantially. Two remaining gaps are worth separating from the fixed bug:

- **citation_treatment (48%)** -- remaining errors are near-misses between semantically adjacent enum values (e.g. human *followed* vs. Phi *discussed*; human *cited* vs. Phi *considered*), not schema violations or invented values -- a genuine judgment-call difficulty on a fine-grained 11-way distinction, not a prompting bug.
- **cited_cases (24% exact-match, recall 0.162)** -- a second, distinct failure mode: on documents with many citations (e.g. **legal_06_328** has 15 gold citations), Phi's list is cut off mid-string (e.g. returning "*bushell v repa*"), consistent with the model hitting its 512-token generation budget before finishing a long list -- a truncation problem, not a comprehension problem.

- Several remaining **visa_category** “failures” are exact-match artifacts rather than content errors -- e.g. Phi answered “*spouse visa*” against a gold label of “*partner/spouse visa (subclass 820)*”: substantively the same category, scored wrong only because the comparison is case-insensitive string equality, not semantic equivalence.

4. Cluster Reading (20%)

K-means (k=6) over the full 220-record corpus produced a low silhouette score (0.018), indicating weak cluster separation -- expected, since every document in this corpus is drawn from the same genre (a Federal Court migration appeal), and full-text TF-IDF-style clustering tends to surface shared procedural boilerplate rather than the topical signal that varies less. Phi's per-cluster accuracy below is scored against the fixed **v2_recovery** prompt (Section 3), not the broken v1_original run, so it reflects the model's real performance rather than the schema-echo bug.

Cluster	Descriptive name	Corpus n	Eval n	Phi acc. (v2)	Top terms
2	Partner/business visa & sponsorship	18	3	0.0	visa, business, relationship, sponsor
4	s501 character-test visa cancellation	24	1	0.0	minister, visa, 501, cancel
0	General tribunal review / FOI-adjacent	18	5	0.2	tribunal, information, review, court
1	General federal appeal (evidentiary)	68	8	0.0	evidence, appellants, federal, appeal
3	General application appeal (procedural)	40	6	0.0	application, decision, appeal, federal
5	General appellant hearing (procedural)	22	2	0.5	appellant, hearing, federal, application

Only two of the six clusters (2 and 4) correspond to a clean, substantively distinct grouping on inspection -- partner/business-visa cases and s501 character-test cancellations, respectively. The other four (0, 1, 3, 5) all read as variations on generic Federal Court migration-appeal procedural language (tribunal, appellant, application, appeal, federal, decision) rather than anything that tracks visa type or case substance; I have given them descriptive names based on their strongest distinguishing terms, but they are honestly closer to one large procedural mass split arbitrarily by k-means than six meaningfully separate topics.

On accuracy: clusters 1 and 3 (evidentiary and procedural appeals) score 0.0, as does cluster 4 (character-test cancellations), while clusters 0 and 5 score 0.2 and 0.5. These figures should be read with caution -- the 25-record evaluation sample is thin per cluster (as few as n=1 for cluster 4 and n=2 for cluster 5), so a single record can swing a cluster's accuracy by 50 percentage points or more. The one directionally interesting pattern is that both substantively coherent clusters (2 and 4) score 0.0, while the highest accuracy (cluster 5, n=2) sits in a generic procedural cluster -- consistent with citation_treatment/cited_cases errors (Section 3) being spread fairly evenly across case types rather than concentrated in a particular visa category, but the sample is too small per cluster to state this as a firm finding.

5. Corpus Limitations

Single source and jurisdiction: every record traces to AustLII via one intermediary (UCI dataset), so the corpus speaks only to Australian Federal Court practice in 2006-2009, not other courts, countries, or eras.

The case-name filter (matching “Minister for Immigration”) is substantively accurate but could miss an unusually styled immigration case. **visa_category** has no ground truth in the source data, so its accuracy numbers reflect agreement between two judgment calls, not validation against a known-correct answer. The corpus records what each case cites and how, not its outcome, so it cannot support questions about how often appeals succeed. The source dataset's license restricts use to research/course purposes with no redistribution, which this project observes by keeping the corpus in a private repository.

6. Conclusion

The initial Human vs. Phi evaluation looked like a near-total model failure (0% accuracy, 88% schema violations), but tracing raw output rather than trusting the aggregate score showed the real cause was a specific, fixable output-format bug -- Phi echoing the JSON Schema envelope instead of returning flat data -- not a comprehension failure. A recovery prompt that named this failure explicitly, rather than generically tightening instructions, eliminated schema violations and recovered substantial accuracy (visa_category 4%→76%, citation_treatment 0%→48%, cited_cases 12%→24%). The cluster analysis adds a complementary read: only 2 of 6 clusters are topically coherent, and per-cluster accuracy against the fixed prompt does not show a strong concentration of errors in any one case type, though the per-cluster evaluation counts are too small to be conclusive on their own.